

ANÁLISIS MATEMÁTICO

$$V_e(s) = R_1 I_1(s) + \frac{1}{Cs} [I_1(s) - I_2(s)]$$

$$\frac{1}{Cs} [I_1(s) - I_2(s)] = (Ls + R_2) I_2(s)$$

$$V_s(s) = R_2 I_2(s)$$

$$V_e(t) = R_1 i_1(t) + \frac{1}{C} \int (i_1(t) - i_2(t)) dt$$

$$\frac{1}{C} \int (i_1(t) - i_2(t)) dt = L \frac{di_2(t)}{dt} + R_2 i_2(t)$$

$$V_s(t) = R_2 i_2(t)$$

$$I_1(s) - I_2(s) = Cs(Ls + R_2) I_2(s)$$

$$I_1(s) = [1 + Cs(Ls + R_2)] I_2(s)$$

$$V_e(s) = R_1 [1 + Cs(Ls + R_2)] I_2(s) + (Ls + R_2) I_2(s)$$

$$V_e(s) = [R_2 + R_1 Cs(Ls + R_2) + Ls + R_2] I_2(s)$$

$$V_e(s) = (R_1 L C s^2 + (L + R_1 R_2 C) s + (R_1 + R_2)) I_2(s)$$

$$\frac{V_s(s)}{V_e(s)} = \frac{R_2}{R_1 L C s^2 + (L + R_1 R_2 C) s + (R_1 + R_2)}$$

Error en estado estacionario

$$e(s) = \lim_{s \rightarrow 0} \left[L - \frac{R_2}{R_1 C s^2 + (L + R_1 R_2 C) s + (R_1 + R_2)} \right]$$

$$e(s) = 1 - \frac{R_2}{R_1 + R_2}$$

$$e(s) = \frac{R_1}{R_2 + R_1}$$

R_1	R_2
1K	1K

$$\left. \vphantom{\frac{R_1}{R_2 + R_1}} \right\} \text{Caso} \rightarrow 1 - \frac{1}{1+1} = 0.5 //$$

R_1	R_2
1K	100K

$$\left. \vphantom{\frac{R_1}{R_2 + R_1}} \right\} \text{Control} \rightarrow 1 - \frac{1}{1+100} = 0.99$$

Estabilidad de sistema

Control = $\lambda_1 = -999989.9898896874$	}	Estable con resp. sobreamortiguada
$\lambda_2 = -1010.0101103126212$		

Caso = $\lambda_1 = -9998.999799939918$	}	//
$\lambda_2 = -2.000200060022004$		